

SEDAR 6: Goliath Grouper and Hogfish

A SEDAR Review Workshop convened to review assessments of Goliath grouper and hogfish snapper. The Goliath grouper assessment was prepared in response to recommendations from the SEDAR 3 (Atlantic Yellowtail snapper) review workshop. The hogfish snapper assessment was prepared by contract with the State of Florida and reviewed by request.

Goliath Grouper

Data Workshop (SEDAR 4)

- The top four prioritized research topics:
 - * Estimation of population size - Estimates of population size were considered to be of highest importance for future management. It was noted that because of the apparent restricted home ranges and high site fidelity, sampling throughout the geographic range would probably be important. Tag/recapture studies were mentioned as a potential monitoring tool.
 - * Demographics - Monitoring the demographics of the population, particularly age composition, could provide valuable information (as it has for red drum in the Gulf of Mexico).
 - * Reproductive Biology - Developing further understanding of the reproductive biology of goliath grouper was considered quite important. Identifying spawning locations, duration and periodicity could be very useful for identifying sites to conduct population surveys.
 - * Historical Abundance - Obtaining information on historical abundance, perhaps via old logbooks, was also considered important.
- Other Research Topics:
 - * It could be very useful to have estimates of unrecorded mortality from accidental or intentional sources, but obtaining such information would be very difficult.
 - * Additional information on stock structure was considered important.
 - * Some thought that it would be useful to have a greater understanding of goliath grouper bioenergetics and trophic relationships. Others asked how that information would assist in a stock assessment.

- * Information identifying the changes in mangrove abundance and distribution, thereby changing available nursery habitat, could assist in developing predictions of future abundance.

Review Workshop

- Estimation of population size: Estimates of population size were considered to be of highest importance for future management. It was noted that because of the apparently restricted home range and high site fidelity characteristic of adults, sampling throughout the geographic range would be important. Tag/recapture studies were mentioned as a potential monitoring tool.
- Estimates of on-going mortality: The issue of ongoing mortality was of critical concern to the Review Panel. Anecdotal information with regard to various sources of this mortality was presented. These sources included longline by-catch, post-release mortality, and illegal harvest. It is extremely important that these sources of ongoing mortality be identified and the magnitude of this mortality estimated.
- Investigations of stock structure: This question was repeatedly raised. The assessment reviewed by the Panel was of necessity limited to south Florida owing to the geographic coverage of the data and the absence of data concerning the stock structure.
- Demographics: Monitoring the demographics of the population, particularly age composition, could provide valuable information.
- Reproductive biology: Developing further understanding of the reproductive biology of Goliath Grouper was considered important.
- Historical abundance and exploitation: Obtaining information on historical abundance was also considered important.
- Survey data. While the Review Panel considered it in the highest degree important to continue the current surveys, it recommended that data collection could be improved by extending survey efforts to better cover the full historical range of the stock.
- The review would have been facilitated if the assessment had been examined by an assessment workshop. It would have been helpful to have the authors of all the relevant documents available to make presentations and answer questions.

Hogfish

- Due to the relatively short time series and relatively low contrasts of CPUE for the available fishery data, the absolute historical limits of stock size and productivity are still somewhat unclear. This would suggest the need for further assessment analyses using

other classes of modeling procedures like stock reduction analyses (Kimura et al. 1984), that could allow the merging of quantitative data time series with observations and opinions about historical states of the fishery.

- Reef-fish commercial log-books should be considered as an additional source of data on commercial catch and effort.
- Weight data, as well as length, should be collected in the head-boat survey;
- Using data from spearfishing tournaments could reinforce length-weight relationships, especially at the right-hand end of the distribution where data are rare.
- The Review Panel considers it important to maintain the current data-collection programs.
- The Review Panel observed that both it, and the presenters, had been handicapped in this review in that neither a data workshop, which would have verified the data sources, nor an assessment workshop had previously been held.

Comments from CIE contractors

(These are excerpted comments intended to highlight suggestions and areas of concern; readers are encouraged to consult the full report for additional details)

- Some guiding documents that would have been useful were not provided. Notable among those were a) brief histories of the assessments, i.e. how they came into being, when, why, and at whose request they were written, and what the prospects would be of having changes made to them; and b) templates for reports—it transpired that the Advisory Report has a fairly specific format that is preferred, and a template or example would have been useful to clarify for the Chairman before the meeting how the information to be derived from the assessments was to be presented in final reports and therefore, to some extent, to define the set of information to be sought.
- A little more information on the meeting format would also have been helpful. The Chairman was not aware before the meeting that the public would be present, and when he was aware of it, he wasn't quite clearly informed what they were doing there and to what degree they were entitled to participate in the process. These questions got sorted out at the meeting, and in the end public participation was in high degree both orderly and helpful.
- Facilities for presenters were minimal and unsatisfactory: they needed more space to put their papers.

- It is a mistake to try to compress such meetings into too few days. Long days put unacceptable pressure on the report-drafting which ideally takes place at the meeting. We had a fairly uncomplicated and trouble-free meeting, but even so did not have much time to spare.
- The delayed response by some Panel members to reports has been a problem; when objections to decisions that were clearly nailed down at the meeting are first voiced two weeks later when the final report is about to be sent, an impossible situation arises in respect of completing and distributing the reports.
- The fact that the (hogfish) assessment had been conducted under contract also proved to be troublesome. The Review Panel was uncertain if the authors could be asked to conduct sensitivity analyses given that they were no longer under contract. It was also unclear who would conduct any subsequent re-assessment.
- For both assessments, the stock area to be assessed was not clearly defined.
- In the report from the Goliath Grouper Data Workshop, distribution was discussed, but more in terms of distribution of the data rather than the species. This was a major issue of discussion for the Review Panel and the lack of a stock definition severely restricted the interpretation of results. For future assessments, this issue should be more closely examined at the Data workshop stage.
- The Peer Review Panel Reports included a section for Stakeholder Comments. This section, independent of and unedited by the Review Panel, provided meeting participants (other than the Review Panel) with a venue to express their views. Given the active participation of certain stakeholders during the workshop, I consider this to be an important and positive feature of these reports.